

Neuron Navigator: a napari Plugin for Interactive Cluster Analysis of Large Numbers of Neuron Reconstructions

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INTRODUCTION

Mouse secondary motor cortex (Allen region MOs) contains neurons with distinct long-range projection patterns. Layer-5 intratelencephalic (IT) neurons project within the telencephalon, including cortex and striatum, whereas pyramidal tract (PT) neurons also innervate targets beyond it, including brainstem nuclei. Hooks et al. (2018) showed that these classes differ in corticostriatal topography, motivating analysis of projection diversity within a cortical territory.⁹

Neuron Navigator is an open-source napari plugin for interactive exploration of CCFv3-registered neuron reconstructions.⁶ Developed using 18,621 published murine cortical neurons,^{1,2} it links anatomical queries, soma-position clustering, and projection-similarity clustering with single-neuron visualization in whole-brain and cortical-flatmap views.

Here we ask whether this workflow can help distinguish IT-like and PT-like projection patterns among MOs neurons. We will select neurons with somata in MOs, cluster them by soma proximity, and re-cluster a spatially compact group by projection similarity outside MOs, measured by Pearson correlations between voxel-wise node counts. Whole-brain inspection will assess whether the resulting groups show anatomy consistent with IT-like or PT-like projections. Cluster membership alone does not establish cell identity.

METHODS

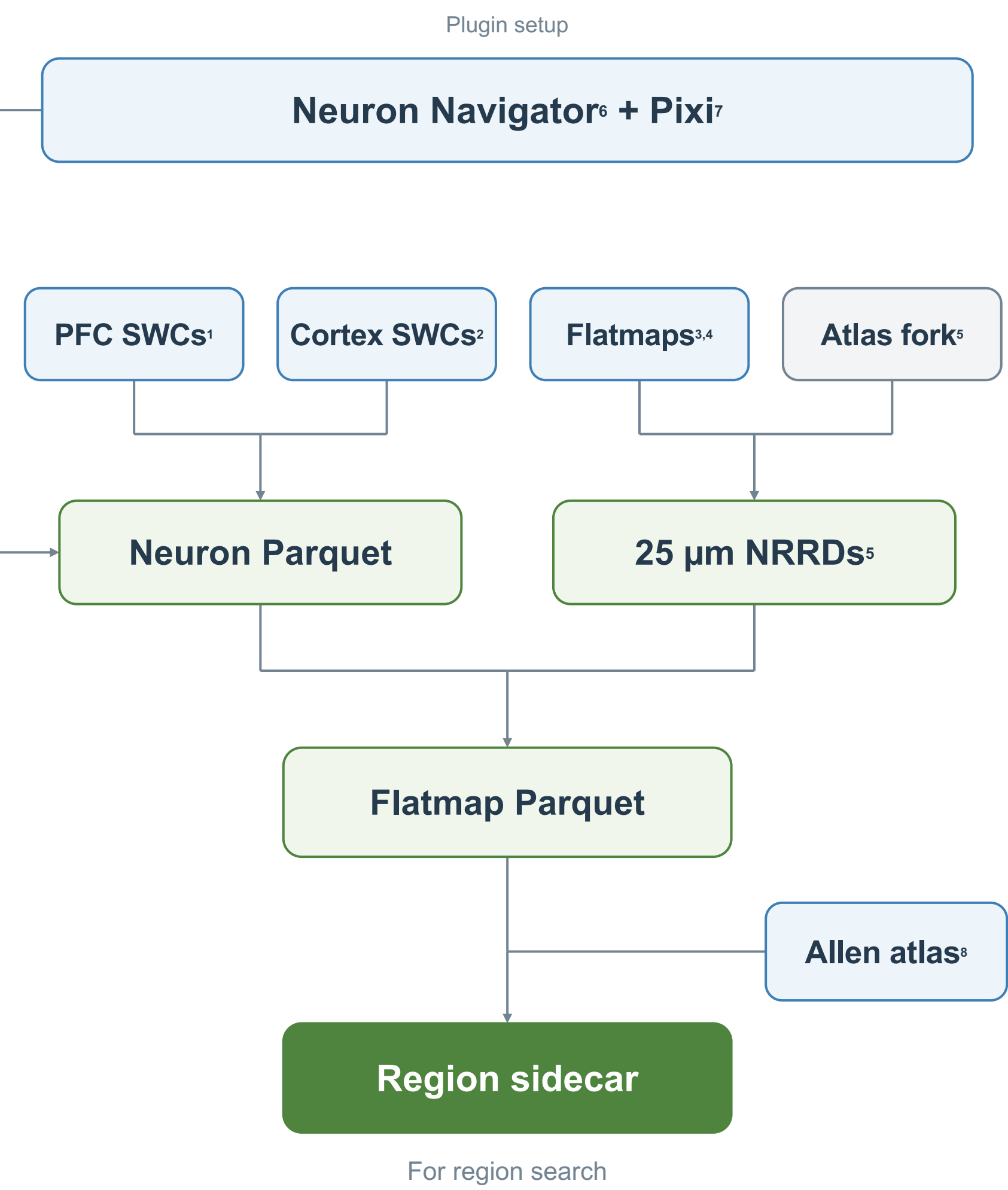


Figure 1. Provenance of the flatmap-enabled neuron Parquet and regional-profile sidecar.

CCFv3-registered SWC reconstructions from the prefrontal cortex and whole-cortex datasets are consolidated into a neuron Parquet using Neuron Navigator. Published cortical flatmap and depth fields are resampled to a 25 μm atlas grid with the atlas-enhancement fork and used to append bilateral flatmap coordinates and cortical depth. Per-neuron summaries derived from Allen atlas annotations are stored in an associated regional-profile Parquet, linked by file_id, to support compound anatomical searches. Arrows indicate preparation dependencies; superscripts identify source references.

TAKEAWAYS

RESULTS

CONCLUSIONS

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